AIMDek Technologies

Introduction to Blockchain

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Agenda

- Introduction to Blockchain
- Blockchain vs cryptocurrency
- Bitcoin vs Ethereum
- Real life and industrial examples
- Business example
- Benefits & challenges
- Q & A

Basic Blockchain terminology

- Block
- Blockchain
- Consensus
- Mining
- Distributed Ledger
- Peer to peer
- Proof of Work

Block

- Block is a set of validated transactions and pointing to next block in the Blockchain or we can term it as container data structure. It is stored at each if the node on network (in case of bitcoin). Each block contains three items:
    - (i) Header
    - (ii) Set of transactions
    - (iii) Hash pointer to previous block
- Header- The header contains metadata about the block. As described by (Cosset, 2017) , there are three different sets of metadata as per Bitcoin Blockchain.
- The hash of previous block – Every block N contains the Hash value of Block N-1.
- Mining competition – This contains the timestamp, nonce and the difficulty. Mining concepts are discussed in more detail consensus algorithms.
- Merkel tree root – This data structure summarizes the transactions in the given block.

Visual Representation of a Block

Block 525252

Block 525253

Timestamp: 27th June, 2018
18:31:00 IST

Transactions: .....

Previous Block reference: 525252

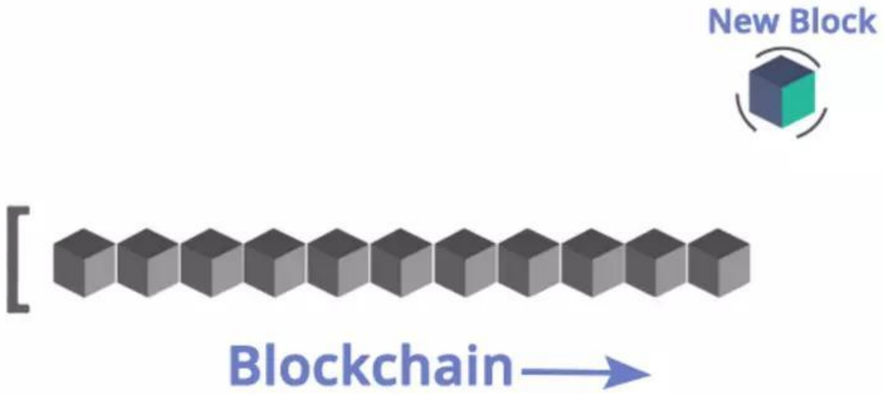
Next block reference: 525254

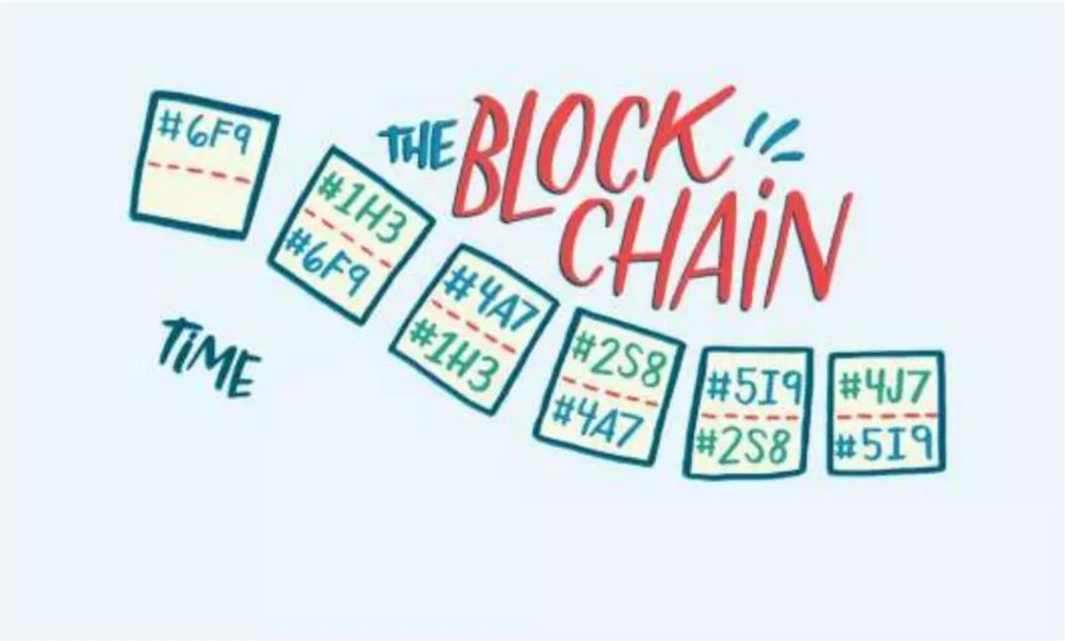
Validation nonce: Hash(T1T2T3T4)

Block 525254

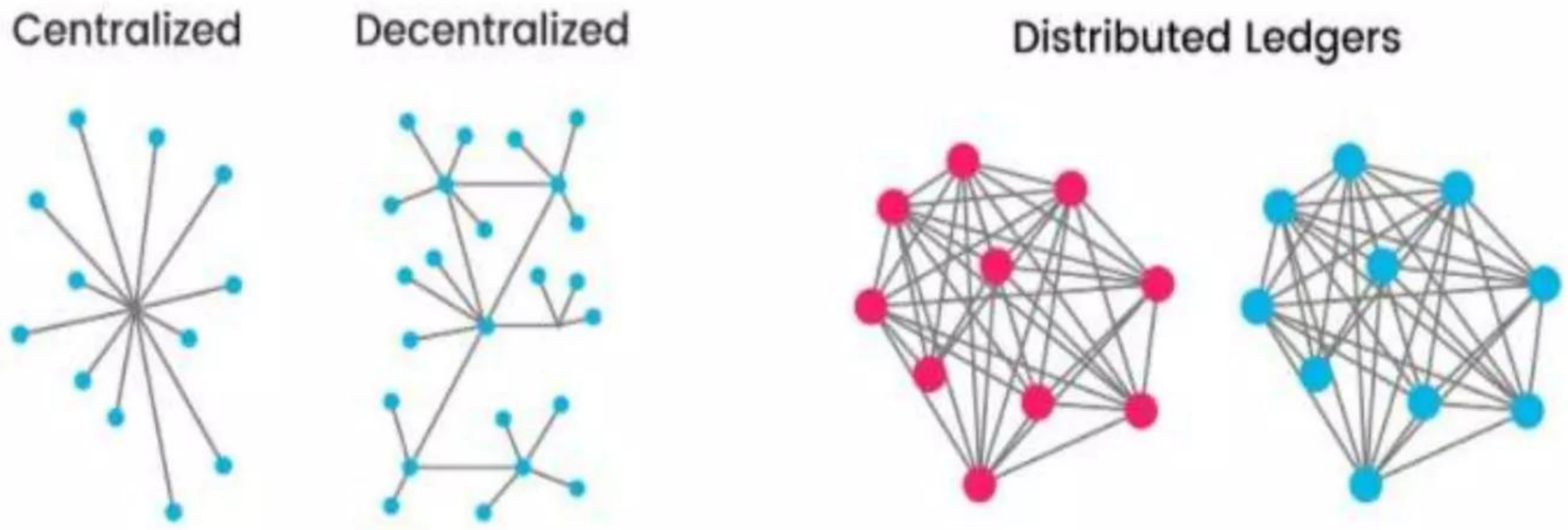
Blockchain

- Blockchain is a distributed, decentralized and immutable database existing on multiple computers at the same time.





Centralized, Decentralized and distributed ledgers



The New Networks

Distributed ledgers can be public or private and vary in their structure and size.

Public blockchains

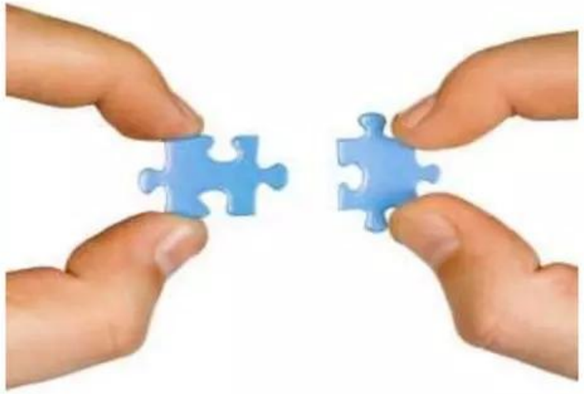
Require computer processing power to confirm transactions (‘ mining’)

- Users (0) are anonymous
- Each user has a copy of the ledger and participates in confirming transaction independently

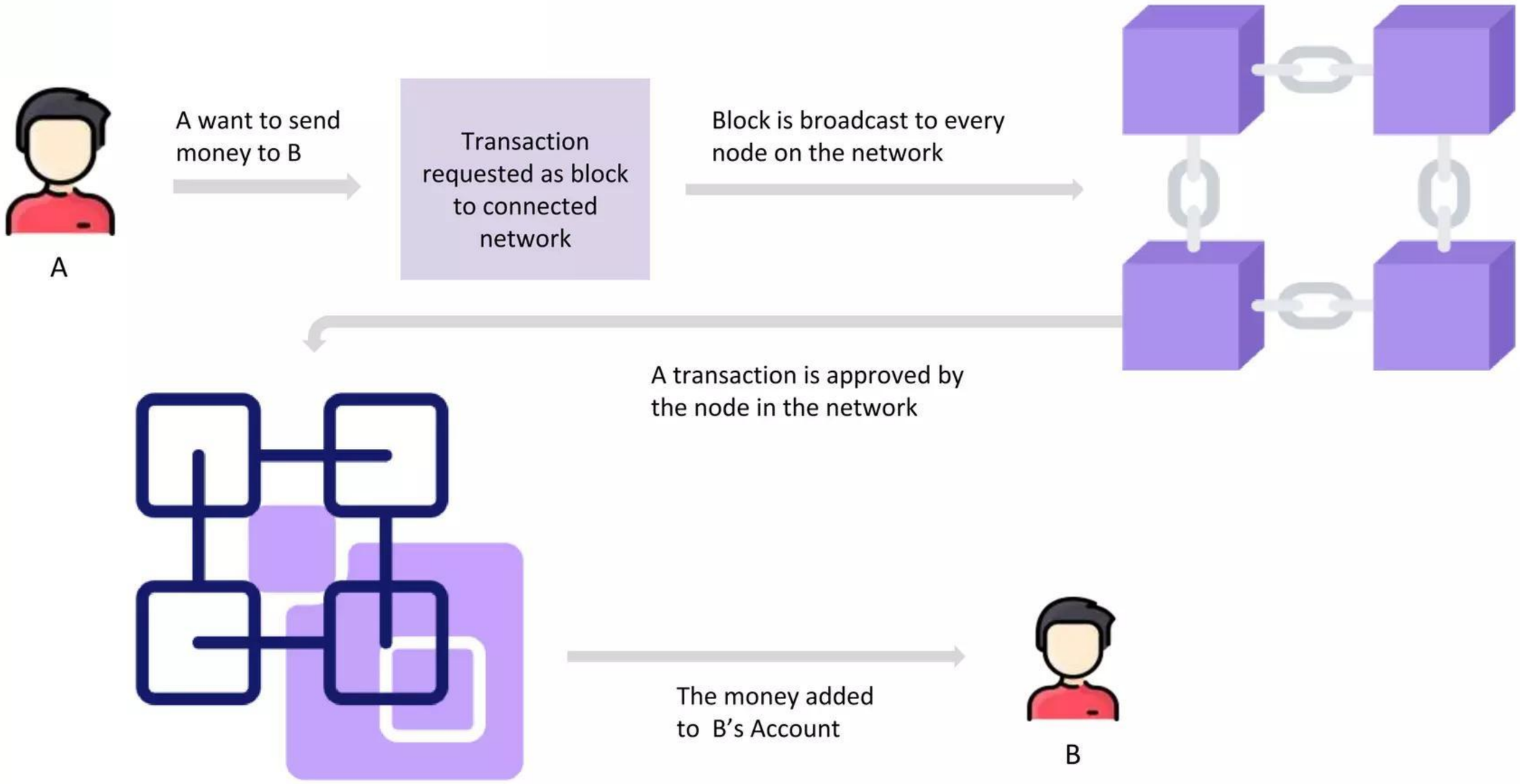
- Users (0) are not anonymous
- Permission is required for users to have a copy of the ledger and participate in confirming transaction

Blockchain vs Cryptocurrency

- Goes hand in hand
- Blockchain technology supports Cryptocurrency
- Bitcoin is best known Cryptocurrency



How Blockchain works



Real life and industrial examples

- **Real time examples:**

  (i) Voting system

- **Industrial examples:**

  (i) Everledger

  (ii) Walmart

  (iii) Microsoft

Business examples

Blockchain use case in Healthcare

Patient consent and Health data management

- Patient control his/her own data
- Secure medical data exchange
- Blockchain ensure consent, compliance, auditability, provenance and governance

Benefits, challenges and pitfalls

Benefits

- Greater Transparency
- JavaScript based
- Decentralized
- Open source & community support
- Adoption of technology

Challenges

- Limited production experience
- Not 100% decentralized
- Availability of developers
- Opinions & Experience

Pitfalls

- New technology - so many of un-supported tools and libraries
- Designing a good blockchain solution
- Customer education

Effects and Future of Blockchain

- Internet of things: Decentralized devices developed by Samsung and IBM
- Ride sharing application: Why to have middle men like Uber and Ola.
- Retail store: Blockchain poses great impact in the online retail industry.
- Indiachain: Tamper proof degree certificate.

Questions and Answers

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